

Dialogue and the Development of Children's Thinking

A sociocultural approach

Neil Mercer and Karen Littleton

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Even casual observations of children's everyday lives reveal that they are constantly engaged in talk and other forms of social interaction. *Dialogue and the Development of Children's Thinking* provides a clear, accessible and well-illustrated case for the importance of spoken dialogue for children's intellectual development during the school years.

The book draws on extensive research to provide a ground-breaking new account of this relationship, and closely relates the research findings to real-life classrooms, so that it is of practical value to teachers and parents concerned that their children are offered the best possible learning opportunities. Importantly, the book provides compelling evidence from original research to show that the quality of classroom talk improves the quality of children's thinking and educational attainment. The book also provides a new and more educationally relevant interpretation of sociocultural theory, based on the work of Vygotsky, which explains the fascinating relationship between engagement in dialogues and learning.

By using evidence of how the collective construction of knowledge is achieved, and how engagement in dialogues shapes children's educational progress and intellectual development, the authors provide a text that will be essential for educational researchers, postgraduate students of education and teachers, and also be of interest to many psychologists and applied linguists.

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LONDON AND NEW YORK

First published 2007
by Routledge
2 Park Square, Milton Park, Abingdon, Oxon OX14 4RN

Simultaneously published in the USA and Canada
by Routledge
270 Madison Ave, New York, NY 10016

This edition published in the Taylor & Francis e-Library, 2007.

“To purchase your own copy of this or any of Taylor & Francis or Routledge’s collection of thousands of eBooks please go to www.eBookstore.tandf.co.uk.”

Routledge is an imprint of the Taylor & Francis Group, an informa business

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British Library Cataloguing in Publication Data
A catalogue record for this book is available from the British Library

Library of Congress Cataloging in Publication Data
A catalog record has been requested for this book

ISBN 0-203-94665-0 Master e-book ISBN

ISBN10: 0-415-40478-9 (hbk)
ISBN10: 0-415-40479-7 (pbk)
ISBN10: 0-203-94665-0 (ebk)

ISBN13: 978-0-415-40478-5 (hbk)
ISBN13: 978-0-415-40479-2 (pbk)
ISBN13: 978-0-203-94665-7 (ebk)

This book is dedicated to the memory of Pam Burns

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Preface

In any area of research, ideas develop through collective as well as individual efforts. This book is the embodiment of the 'interthinking' and collaborative work of a team of researchers, teachers and advisers who have, over 16 or so years, been committed to finding theoretically informed ways of helping children to think together. The ideas presented here have been collectively constructed by the *Thinking Together* team. The other principal members of this team have been Lyn Dawes and Rupert Wegerif, but important contributions to *Thinking Together* have also been made by Juan Manuel Fernández Cardenas, Jan English, Eunice Fisher, Jonathan Giles, Steve Higgins, Ruth Holmwood, Jenny Houssart, Judith Kleine Staarman, Tara Lovelock, Frank Monaghan, Sylvia Rojas-Drummond, Denise Rowe and Claire Sams. Pam Burns provided excellent secretarial support for the research. Many other people have also been important in shaping what we have written here, and of these we would particularly like to acknowledge the influence of Robin Alexander, Jaume Ametller, Douglas Barnes, Caroline Coffin, Mariette de Haan, Derek Edwards, Ed Elbers, Futoshi Hiruma, Paul Light, Janet Maybin, Dorothy Miell, Andy Northedge, Ingvill Rasmussen, Roger Säljo, Phil Scott, Joan Swann and Stephanie Taylor. Without the expert help of Joan Dearman and Carol Johns-Mackenzie, producing the text of this book would have been much more difficult, and excellent editorial support at Routledge has ensured that the process of publication ran smoothly. Thanks are particularly due to Lyn Dawes and Judith Kleine Staarman, who provided constructively critical comments on drafts of the manuscript, and to David Shaw, who proof-read the final version.

We wish to gratefully acknowledge the financial support for research provided by the Economic and Social Research Council (C00232236, R000232731, R000221868 and RES-000-23-0939), The Nuffield Foundation (EDU/00169/G), The Esmeé Fairbairn Foundation, Milton Keynes Council, The British Council and (in Mexico) the Consejo Nacional de Ciencia y Tecnología (CONACYT) and the Dirección General de Asuntos del Personal Académico of the Universidad Nacional Autónoma de México (DGAPA-UNAM). Our thanks also go to all the teachers and children who have participated in our research. (In order to maintain confidentiality, in the transcribed examples the names of all participants have been changed.)

In writing this book we have drawn on our body of previously published research and teaching materials. Some of the material in Chapter 7 is also included in an article by Neil Mercer entitled, 'The seeds of time: Why classroom dialogue needs a temporal analysis', in the *Journal of Learning Sciences*, volume 16.

Figure 2.1 is reprinted with permission from Taylor and Francis. This figure originally appeared in D.Faulkner, K.Littleton and M.Woodhead (1998) *Learning Relationships in the Classroom*, London: Routledge.

Figure 4.1 is reproduced with the kind permission of Phil Scott.

Appendix A is reprinted with permission from Taylor and Francis. The material originally appeared in L.Dawes and C.Sams (2004) *Talk Box: Speaking and Listening Activities for Learning at Key Stage 1*, London: David Fulton.

Notes on transcription

We have used a very simple transcription format, in which speech is rendered as grammatical phrases and sentences, to represent the sense that we, as researchers with access to the raw data, made of what was said. Information about equipment, non-verbal aspects of communication and any other contextual information that we considered pertinent to the analysis is to be found in italics in a third column or in parentheses. Our judgement was that the inclusion of additional information at our disposal, such as length of pauses or other prosodic and contextual details, would be distracting to readers and irrelevant to the issues we are addressing. Non-word utterances such as ‘mm’/‘ooh’ are included when they are judged to have a communicative function (for example to show surprise, agreement, or to extend a speaker’s turn in the face of possible interruptions). Words spoken emphatically are in italics. Simultaneous speech is shown by the use of brackets (I) preceding each utterance. Where the accurate transcription of a word is in doubt, it is in parentheses. When utterances could not be heard or deciphered, we say so.

Chapter 1

Why dialogue?

Our main aim in this book is to explain how classroom dialogue contributes to children's intellectual development and their educational attainment. To do so, we will make use of the results of research we and our colleagues have carried out in schools over the last 16 or so years, as well as the work of many other researchers. Examples drawn from interactions in classrooms will illustrate our discussion of how children develop as thinkers, problem-solvers and effective members of collective endeavours, and of how teachers contribute to that development. But we will do more than make a case for the importance of dialogue as the prime tool for helping children achieve an education from their school experience. We will also describe how talk in classrooms can be analysed in terms of its functions and quality, making clear the implications of this analysis for the practice of teaching and learning. What is now known about the psychological functions of dialogue is not only relevant to the academic study of cognitive development and learning: it is also of practical value to people such as teachers and parents who are concerned with ensuring that children are offered the best educational opportunities. Our hope is that what we have written here will inform practical action.

What kinds of dialogue are we interested in?

'Dialogue' is sometimes used in a broad sense to mean the interchange of ideas between one source and another. It is used to refer to such different processes as an individual reader grappling with the ideas in a book and negotiations between social groups or organizations. Such abstract or metaphorical uses are not wrong or without value, but here we mean something more concrete and specific. In this book, the kind of dialogue we are interested in is classroom talk. To narrow the focus even more, we will concentrate on talk that takes place in the course of educational activities. Though we recognize the importance in educational activity of reading and writing, the use of gesture, diagrams and other non-verbal ways of interacting, our view is that the distinctive role of spoken language in learning and development justifies it being given attention in its own right. Moreover, although there has been a good deal of research on classroom talk, we do not think that enough attention has been given to the relationship between the quality of talk and learning outcomes. The dialogues we will consider include teacher-student exchanges and discussions amongst students. Both those types of dialogue have potential value for learning and development, but we will show that each has special functions.

Why does classroom dialogue deserve more attention?

There are many ways that people can make sense of the world together, using actions, graphic representations and various kinds of symbol systems as well as language. All of these will probably influence the ways individuals come to make sense of the world on their own. We would never claim that everything that can be thought can be thought in language, or that language is involved in all rational thinking. But language is without doubt the most ubiquitous, flexible and creative of the meaning-making tools available, and it is the one most intimately connected to the creation and pursuit of reasoned argument. Becoming an educated person necessarily involves learning some special ways of using language: and language is also a teacher's main pedagogic tool. For these reasons language, and especially spoken dialogue, deserves some special attention.

Understanding the role of spoken dialogue in learning and development must involve consideration of children as social actors, and not just as developing individuals. Social experience does not provide all children with the same language experiences, so we cannot assume that all children naturally have access to the same opportunities for developing their use of language as a tool for learning, reasoning and solving problems. This is nothing to do with the obvious differences of mother tongue, dialect and accent, which reflect children's social origins, and which tend to figure in popular debates about the quality of children's talk (and what schools should do about it). Within the social differentiation that is typical within any society today, children's language experiences may vary in other ways that, while subtle, are potentially of greater significance for their educational progress. Research in the USA has provided evidence to support the view first advanced by Bernstein (1975) and others some years ago, that the amount and quality of the dialogue children experience at home is strongly correlated with their eventual academic attainment (Hart and Risley, 1995). Although life will provide most children with a rich and varied language experience, in some homes rational debates, logical deductions, reflective analyses, extended narratives and detailed explanations may never be heard. How can children be expected to incorporate such ways of using language into their repertoires, if they have no models for doing so? Without the example and guidance of a teacher, many children may not gain access to some very useful ways of using language as a tool for reasoning, learning and working collaboratively because those 'ways with words' (Heath, 1983) are simply not part of their experience. As we will describe in Chapter 6, our own research has shown that when teachers focus on the development of children's language as a tool for reasoning, this can lead to significant improvements in the quality of children's problem solving and academic attainment. Nevertheless, relatively few schools explicitly teach this kind of language use. One of the arguments we will make in the book is that there is not enough emphasis in educational policy and practice on the value of teaching children how to use language for learning.

What do we mean by learning and development?

‘Learning’ and ‘development’ are terms that have both been used a great deal in developmental psychology, while ‘learning’, often in the company of ‘teaching’, is of course a common term in educational studies. We have used the two words together because we feel that both are required to invoke the kinds of cognitive, intellectual changes that we are interested in here. Both have been applied to individuals, to groups and to societies as a whole. The educational researcher Watkins (2003) distinguishes three influential conceptions of learning: ‘Learning is being taught’, ‘Learning is individual sense-making’, and ‘Learning is building knowledge with others’. These are not at all incompatible, and from our perspective are complementary. ‘Learning’ is normally associated with the gaining of knowledge, with the acquisition of some fact or skill. We can learn something, and we may forget it. In this book we are not concerned with the processes by which people commit new information to memory (as when one memorizes a friend’s new telephone number, or when a science student becomes able to recall the sequence of elements in the periodic table). Rather, we are concerned with the ways people learn to make sense of the world, become able to solve problems and—in school settings—take on new perspectives such as those inherent in science, mathematics and other subjects.

‘Development’ usually implies some change of a progressive kind. It invokes ideas of some sort of growth, the emergence of a new entity, the arrival of a new state of affairs. Children often forget something they have learned, and in many circumstances this would not be a cause for alarm; but developmental regression in a child would be expected to ring alarm bells. We are using both terms because we are not only interested in how dialogue contributes to the ways that people make sense together and gain knowledge from social interaction, but also how it enables them to become progressively more able to carry out certain kinds of intellectual activities.

How does interaction help learning and development?

Research into the processes of learning and cognitive development has been transformed in the last 20 years by the emergence of sociocultural theory, which is also sometimes described as ‘socio-historical’ and ‘cultural-historical’ (see, for example, Wertsch, 1985a; Daniels, 2001; Wells and Claxton, 2002). Its origins lie mainly in the work of the Russian psychologist Lev Vygotsky (for example, Vygotsky, 1962, 1978). Sociocultural research is not a unified field, but those within it treat communication, thinking and learning as processes shaped by culture, whereby knowledge is shared and understandings are jointly constructed. Communicative events are shaped by cultural and historical factors, and thinking, learning and development cannot be understood without taking account of the intrinsically social and communicative nature of human life.

From a sociocultural perspective, humans are seen as creatures who have a unique capacity for communication and whose lives are normally led within groups, communities and societies based on shared ways of using language, ways of thinking,

social practices and tools for getting things done. Education is seen as a dialogic process, with students and teachers working within settings that reflect the values and social practices of schools as cultural institutions. A sociocultural perspective raises the possibility that educational success and failure may be explained by the quality of educational dialogue, rather than simply by considering the capability of individual students or the skill of their teachers. It encourages the investigation of the relationship between language and thinking and also of the relationship between what Vygotsky called the 'intermental' and the 'intramental'—the social and the psychological—in the processes of learning, development and intellectual endeavour. Partly through the influence of these ideas, social interaction has increasingly come to be seen as significant in shaping children's cognitive development. We give special attention to this topic in Chapter 2.

How does language enable collaborative learning?

Many human activities involve not just the sharing of information and the coordination of social interaction, but also people working together to solve problems. When they do so, people do not only interact, they 'interthink', combining their intellects in creative ways that may achieve more than the sum of the parts. In such problem-solving situations there is a dynamic engagement with ideas amongst partners, with language as the principal means for establishing shared understanding, testing out possible solutions and trying to reach some agreement. Thinking together is an important part of life, but it has traditionally been ignored or even repressed in school. In recent years, though, the potential value of children's collaborative activity for their learning and development has begun to be appreciated. In Chapter 3 we review research on children's collaborative learning for what it can tell us about the factors that contribute to joint activity being productive.

How can dialogue with a teacher help children learn?

In Chapter 4, we turn our attention to talk between teachers and students. This is a topic that has been studied by many people for many years, but new and useful insights have emerged in relatively recent times. The findings of research now offer a clearer and more secure understanding of how teacher-student dialogue can be used to good effect—and of how opportunities for productive dialogue may sometimes be squandered. Our own interest in such matters is not merely as detached observers, analysts, assessors or critics of the education system. As members of an international team of applied educational researchers, we have worked closely with teachers and children to try to understand and improve the educational process.

Discussions about the teacher's role sometimes oppose two models: the instructor and transmitter of knowledge on the one hand, and on the other the facilitator and co-learner. These models often figure in debates about the relative value of traditional and progressive educational methods, or about transferring control of education from the teacher to the learner. Such simple dichotomies are useful for political rhetoric, but are of

little value for understanding what makes education more or less effective. Instead, we need to appreciate the diverse ways that the roles of teachers and learners can be enacted and to understand that they can shift and adapt within the classroom context, from one activity to another as is appropriate. As we will show, the close study of the dialogues between teachers and students can help the planning of activities to ensure that opportunities are provided for teachers and students to construct knowledge and understanding together.

What are the implications for educational theory and practice?

Vygotsky proposed that children's intellectual development is shaped by the acquisition of language, because language makes dialogue possible between children and other members of their community. This proposal is fundamental to sociocultural explanations of learning and development. But our allegiance to a sociocultural perspective does not mean that we have simply assumed the truth of Vygotsky's proposal—for which he provided little empirical evidence. One of the aims of the research we and our colleagues have carried out has been to put Vygotsky's ideas to the test. We have done so through creating and evaluating interventional educational programmes that embody a distinctive approach to language as a tool for teaching and learning. This approach, called *Thinking Together* and which is described in Chapter 5, has been tried and tested in schools in several countries. It represents sociocultural theory in action, in two rather different ways. First, it illustrates the practical consequences of basing an approach to teaching upon the notion of language as the principal cultural and psychological tool for building knowledge. Second, its implementation has allowed ordinary classrooms to become environments for testing the utility of sociocultural theory. As we will show in Chapter 6, the results have not only provided support for Vygotsky's claims about the relationship between the social and the psychological, they also have implications for educational practice.

What are the implications for future research?

A key feature of our approach to analysing and evaluating educational dialogue is to treat it as a process that is orientated to both the past and the future. That is, the meaning and value of dialogue for those involved in it depends on its history and where it is heading. This is not just a matter of the interests of speakers being brought to bear and so influencing the direction of the dialogue, but also the extent to which there is a basis of common knowledge that will enable participants to make sense together. Dialogues are cultural artefacts, because they embody participants' practical knowledge about how to talk in a particular kind of situation. Classroom dialogue depends on speakers understanding the 'rules of the game'. However, as we explain in Chapters 4, 5 and 6, those rules may not always be ones that help children gain an education from their school experience.

In any particular interaction, speakers use their existing knowledge to build contextual foundations for the progress of their talk. They use talk itself as a tool for creating new shared understanding. If we are interested in explaining and evaluating the processes of teaching and learning, we need to understand how teachers and learners use language to create new common knowledge. But this is a difficult topic to research, because spoken interaction, in classrooms or any other settings, has a temporal dimension. School lessons begin and end, but the dialogues within them do not necessarily have the same beginnings and ends. People take up topics they began to discuss on earlier occasions, refer to events that have happened in the meantime, and often orientate their discussions towards future activities and outcomes. Educational dialogues are particularly dependent on appeals to knowledge already gained and to learning goals ahead. In Chapter 7 we look at this vital, intriguing but complicated aspect of talk in classrooms and its significance for investigating and understanding learning and development.

In the final chapter of the book, we will do what is customary at that point: summarize what is known about the nature and functions of educational dialogue and draw some conclusions about how this should affect our understanding of children's learning and intellectual development. We will also draw out some implications for the practice of teaching and learning.

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